1 Intersections and Unions of Subspaces

In set theory, one can form new subsets from old ones by taking intersections and unions. In linear algebra, we are interested in subspaces of a vector space, not arbitrary subsets. Let $\mathscr{S}(V)$ denote the set of all subspaces of a vector space V.

Proposition 1.1. *The intersection of any collection of subspaces of V is a subspace of V.*

Proof. Let $\{W_\alpha\}_{\alpha\in\mathscr{A}} \subset \mathscr{S}(V)$ be an arbitrary collection of subspaces of V. Since each W_α is a subspace of V, we have $0 \in W_\alpha$ for each $\alpha \in \mathscr{A}$, and therefore $0 \in \bigcap_{\alpha\in\mathscr{A}} W_\alpha$. The rest of the proof is left as an exercise. □

Exercise 1.2. Complete the proof of Proposition 1.1.

While Proposition 1.1 shows that the intersection of subspaces of V is always a subspace of V, the union of subspaces is rarely a subspace.

Example 1.3. We have seen that every line through the origin in $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$. Consider the union of the x-axes and y-axes. Then $(1,0)$ and $(0,1)$ are both in the union, but $(1,0)+(0,1) = (1,1)$ is not. Since the union of the x-axes and y-axes is not closed under addition, it is not a subspace of $\mathbb{R}^2$.

Exercise 1.4. Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other. (*Hint*: To prove that $U \cup W$ is not a subspace if neither U nor W is contained in the other, follow the same strategy as in Example 1.3: choose $u \in U - W$ and $w = W - U$ and consider $u + w$.)